INCH-POUND

MIL-S-19500/522(ER)
AMENDMENT 5
14 February 1989
SUPERSEDING
AMENDMENT 4
28 August 1978

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, HIGH-FREQUENCY
DEVICE TYPES: JAN 2N6603 AND TXV 2N6603, JAN 2N6604 AND TXV 2N6604

This amendment forms a part of MIL-S-19500/552(ER) dated 21 August 1976, and is approved for use by the Laboratory Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

The document identification date shall be changed to read: 3 September 1976 in place of 21 August 1976.

1.3, delete and substitute:

	P_T				V_{CBO}	V_{CEO}	V_{EBO}	I_C	T_j	T_{stg}
	1/ T_A = 25°C	2/ T_C = 125°C	3/ T_A = 25°C	4/ T_C = 125°C						
	mW	mW	mW	mW	V dc	V dc	V dc	mA dc	°C	°C
2N6603	300	400	---	---	25	15	3.0	30	200	-65 to +200
2N6604	---	---	350	500	25	15	3.0	30	200	

1/ Derate linearly 1.71 mW/°C for T_A > 25°C.
2/ Derate linearly 5.33 mW/°C for T_C > 125°C.
3/ Derate linearly 2.00 mW/°C for T_A > 25°C.
4/ Derate linearly 6.66 mW/°C for T_C > 125°C.

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1.4, delete and substitute:

"

	h_{FE}				C_{cb}		NF		G_{pe}			
	V_{CE} = 10 V dc, I_C = 15 mA dc		V_{CE} = 10 V dc, I_C = 30 mA dc		V_{CB} = 10 V dc, I_E = 0, .1 MHz ≤ f ≤ 1 Hz		V_{CE} = 10 V dc, I_C = 5 mA dc, f = 1 GHz		V_{CE} = 10 V dc, I_C = 15 mA dc, f = 1 GHz		V_{CE} = 10 V dc, I_C = 30 mA dc, f = 1 GHz	
	MIN	MAX	MIN	MAX	MIN	MAX	MIN	MAX	MIN	MAX	MIN	MAX
	---	---	---	---	P_F	P_F	db	db	db	db	db	db
2N6603	30	200	---	---	.25	.75	1.0	2.5	15	21	---	---
2N6604	---	---	30	200	.30	.80	1.5	3.0	---	---	15	21

"

3.5, delete and substitute the following paragraph:

"3.5 Marking. All device marking specified in MIL-S-19500 may be omitted from the body of the transistor except the applicable prefix and device type number; however, all marking specified in MIL-S-19500 shall be marked on the device carrier except country of orgin and manufacturer's identification. Examples of acceptable marking are as follows:

J6 JV6
603 603 "

PAGE 4

4.4.2, third line: Delete the word "timers" and substitute the word "tuners".

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4.6.8: Delete "P_T = 400 mW" and "P_T = 500 mW" and substitute "P_T = 300 mW" and "P_T = 350 mW" respectively.

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TABLE I, Subgroup 2, Breakdown voltage emitter to base test, limits column: Delete "2 V dc Min" and substitute "3 V dc Min".

TABLE I, Subgroup 3, Power-gain, limits column: Delete "14-20 db" and substitute "15-21 db".

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TABLE I, Subgroup 3, Collector-base capacitance, limits column for 2N6604 only: Delete ".45-.95 P_F" and substitute ".30-.80 P_F".

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TABLE II, Subgroup 2, End points, power-gain test, limits column: Delete "14-20 db" and substitute "15-21 db".

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TABLE II, Subgroup 4, Examination column: Delete "(lead fatigue)" and substitute "(lead tension)".

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TABLE II, Subgroup 5, End points, power-gain test, limits column: Delete "14-20 db" and substitute "15-21 db".

TABLE II, Subgroup 6, Steady-state operation life, details column: Delete "P_T = 400 mW" and "P_T = 500 mW" and substitute "P_T = 300 mW" and "P_T = 350 mW" respectively.

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TABLE III, Subgroup 5, Steady-state operation life, details column: Delete "P_T = 400 mW" and "P_T = 500 mW" and substitute "P_T = 300 mW" and "P_T = 350 mW" respectively.

TABLE III, Subgroup 5, End points: Delete "(same as Subgroup 5 inspection)" and substitute "(same as subgroup 5 of group B inspection)".

TABLE III, Subgroup 7, Input reflection coefficient test, limits column, 2N6604 only: Delete ".6 min .9 max" and "+160 min -170 max" and substitute ".45 min .75 max" and "+155 min -175 max".

TABLE III, Subgroup 7, Output reflection coefficient test, limits column, 2N6603 only: Delete ".3 min .5 max" and "-55 min -25 max" and substitute ".25 min .5 max" and "-70 min -40 max".

* TABLE III, Subgroup 7, Output reflection coefficient test, maximum limits column, 2N6603: Delete "-40 (added from a previous amendment)" and substitute "-30".

TABLE III, Subgroup 7, Output reflection coefficient test, limits column, 2N6604 only: Delete ".1 min .3 max" and "-95 min -65 max" and substitute ".2 min .4 max" and "-70 min -40 max".

* TABLE III, Subgroup 7, Output reflection coefficient test, maximum limits column, 2N6604: Delete "-40 (added from a previous amendment)" and substitute "-30".

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5.1, delete and substitute:

"5.1 see MIL-S-19500, section 5, except that electrostatic packaging as specified in paragraph 3.1.4.1 of MIL-S-19491E is not required."

NOTE: The margin of this specification is marked with an asterisk to indicate where changes from the previous issue were made. This was done as a convenience only, and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

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CONCLUDING MATERIAL

Custodians:
Army - ER

Preparing activity:
Army - ER

Agent:
DLA - ES

(Project 5961-A959)

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